

An explicit dependence among the colored partition maps at $n = 2$

Counterexample packet for arXiv:1011.5400

11 August 2026

Status. The linear-independence clause of Theorem 3.5 in arXiv:1011.5400 is false as printed. For $s = 5$, $n = 2$, and $(k, l) = (0, 6)$, eleven distinct diagrams in $D_5(0, 6)$ satisfy the explicit nonzero linear relation below. In fact the same relation belongs to $D_s(0, 6)$ for every $s \geq 3$.

This does not challenge the theorem's spanning statement. It shows that the argument counting diagrams as a basis, and conclusions using that count at $n = 2$, require a small-rank correction.

The source claim and Gram question

The source works on $H = \mathbb{C}^{2n}$ and assigns a map $T_\pi : H^{\otimes k} \rightarrow H^{\otimes l}$ to each admissibly bicolored noncrossing partition $\pi \in D_s(k, l)$. Source PDF page 10 states:

Theorem 3.5. *For the quantum group $G^+(\widehat{\mathbb{Z}_s^{*n}})$, with $5 \leq s < \infty$ and $n \geq 2$, we have:*

$$\text{Hom}(u^{\otimes k}, u^{\otimes l}) = \text{span}(T_\pi | \pi \in D_s(k, l))$$

In addition, the linear maps T_π appearing on the right are linearly independent.

After the proof, source PDF page 12 asks for the Gram determinant of these maps and suggests proving independence by showing that determinant is nonzero. The counterexample occurs at the first degree where the relevant determinant has a factor $2n - 4$.

Notation for the eleven diagrams

Put

$$X = \{1, 2\} \times \{0, 1\}, \quad \overline{(i, a)} = (i, 1 - a), \quad H = \mathbb{C}^X.$$

Write $c_0 = wb$ and $c_1 = bw$. For a colored partition π of six lower points, set $\xi_\pi = T_\pi(1) \in H^{\otimes 6}$. The source's definition says that a tensor coordinate of ξ_π is one exactly when, in every block, all black indices equal some $I \in X$ and all white indices equal $\bar{I}$.

We specify a block by its increasing list of positions followed by its color word. For $a, b \in \{0, 1\}$ define:

$$\begin{aligned} A_{ab} &= \{123456[c_0c_ac_b]\}, \\ B_{12}^a &= \{1234[c_0c_a]\}\{56[c_0]\}, \\ B_{13}^a &= \{1256[c_0c_a]\}\{34[c_0]\}, \\ B_{23}^a &= \{12[c_0]\}\{3456[c_0c_a]\}, \\ C &= \{12[c_0]\}\{34[c_0]\}\{56[c_0]\}. \end{aligned}$$

For example, A_{10} has color word $wbbwwb$, while B_{13}^1 has the outer block $\{1, 2, 5, 6\}$ colored $wbbw$ and the inner block $\{3, 4\}$ colored wb .

Every displayed partition is noncrossing. Every block contains equally many black and white legs, so the defining congruence holds for every s ; the eleven diagrams are distinct even after the source's blockwise color-swap identification.

Counterexample theorem. *On $H = \mathbb{C}^4$ one has*

$$2 \sum_{a,b \in \{0,1\}} \xi_{A_{ab}} - \sum_{a \in \{0,1\}} (\xi_{B_{12}^a} + \xi_{B_{13}^a} + \xi_{B_{23}^a}) + \xi_C = 0. \quad (1)$$

Consequently the maps indexed by $D_5(0, 6)$ are not linearly independent when $n = 2$.

Proof intuition. All eleven tensors live on coordinates in which each adjacent pair has opposite colors. Such a coordinate is determined by three indices $x_1, x_2, x_3 \in X$. Joining two pair-blocks forgets their second coordinates and tests only whether their first coordinates in $\{1, 2\}$ agree. Joining all three tests whether all three first coordinates agree. The relation is then just the pigeonhole fact that among three elements of a two-point set, either all three are equal or exactly one pair is equal.

Proof. Fix a basis coordinate $(I_1, \dots, I_6) \in X^6$. Every term in (1) vanishes unless

$$I_2 = \bar{I}_1, \quad I_4 = \bar{I}_3, \quad I_6 = \bar{I}_5. \quad (2)$$

Assume (2), put $x_j = I_{2j-1}$, and let $p_j \in \{1, 2\}$ be the first coordinate of x_j .

For a fixed pair of pair-blocks, summing the two orientations c_0, c_1 gives the indicator that their first coordinates agree. Thus

$$\sum_a \xi_{B_{12}^a} = \mathbf{1}_{p_1=p_2}, \quad \sum_a \xi_{B_{13}^a} = \mathbf{1}_{p_1=p_3}, \quad \sum_a \xi_{B_{23}^a} = \mathbf{1}_{p_2=p_3}$$

at the chosen tensor coordinate. Similarly, $\sum_{a,b} \xi_{A_{ab}} = \mathbf{1}_{p_1=p_2=p_3}$, while $\xi_C = 1$. The coefficient of that coordinate on the left of (1) is therefore

$$2\mathbf{1}_{p_1=p_2=p_3} - \mathbf{1}_{p_1=p_2} - \mathbf{1}_{p_1=p_3} - \mathbf{1}_{p_2=p_3} + 1. \quad (3)$$

If all three p_j are equal, (3) is $2 - 3 + 1 = 0$. Otherwise, because $p_j \in \{1, 2\}$, exactly one pair is equal, and it is $0 - 1 + 1 = 0$. Hence every tensor coordinate vanishes, proving (1). Its eleven coefficients are not all zero, so it is a genuine dependence among distinct elements of $D_5(0, 6)$. $\square$

Exact Gram audit

Let $N = 2n$. An independent signed-union-find enumeration constructs all 33 diagrams in $D_5(0, 6)$ and their exact polynomial Gram matrix. It gives

$$\det G_{5,6}(N) = N^{33}(N-2)^{26}(N-4)^2. \quad (4)$$

At $n = 2$, the 33×33 matrix has exact rank 31. Thus the hand relation is corroborated by the full Gram computation, which also provides a concrete low-degree answer to the source's determinant question. The proof of the counterexample itself does not depend on the enumeration.

The exhaustive coordinate checker verifies (1) on all $4^6 = 4096$ basis coordinates and confirms that every diagram occurs nontrivially. Both exact programs are included in the packet.

Literature and scope. Exact-title, theorem-phrase, determinant, colored-partition, and citation searches found later classification work but no published correction or this dependence through 11 August 2026. The same authors' arXiv:1109.6184 answers the source's separate geometric question about K_n^+ ; it does not address this rank defect. No claim is made here about independence for $n \geq 3$ or about a closed determinant formula in all degrees.

Human review. Check that the block/color notation matches Definition 3.1 of the source, that the eleven partitions are distinct elements of $D_5(0, 6)$, and that the two-orientation sum tests equality of the first coordinates. These checks are finite and independent of quantum-group machinery.

References

- [1] T. Banica and A. Skalski, *Quantum isometry groups of duals of free powers of cyclic groups*, arXiv:1011.5400; International Mathematics Research Notices 2012, no. 9, 2094–2122. See Definition 3.1, Theorem 3.5, and the Gram question on source PDF pages 9–12.
- [2] T. Banica and A. Skalski, *Quantum symmetry groups of C^* -algebras equipped with orthogonal filtrations*, arXiv:1109.6184. This supplies the later block-length interpretation of K_n^+ mentioned only for scope separation.